



A three-dimensional (3D) analytical solution for the ultimate side shear resistance of rock-socketed shafts

Haohua Chen^a, Hehua Zhu^b, Lianyang Zhang^{a,*}

^a Department of Civil and Architectural Engineering and Mechanics, The University of Arizona, Tucson, 85719, Arizona, USA

^b Department of Geotechnical Engineering, Tongji University, Shanghai, 200092, China

ARTICLE INFO

Keywords:

Rock socketed shafts
Ultimate side shear resistance
Disturbance
3D strength
Socket roughness

ABSTRACT

This paper presents a new analytical solution to the ultimate side shear resistance τ_{sf} of rock socketed shafts with consideration of three-dimensional (3D) strength and shaft sidewall roughness. Specifically, the shaft sidewall profile is simplified as triangular asperities and micro-mechanical analysis is performed on the shear behavior of one representative asperity. A 3D version of the Hoek-Brown (HB) criterion for rock mass and the equilibrium equations in a cylindrical coordinate system are first utilized to derive the governing equations. Then the governing equations are solved by using the method of characteristics. Finally, the τ_{sf} is obtained by considering the constant normal stiffness condition and the relative displacement of the rock mass due to sliding along the shaft-rock mass interface. Twenty-four (24) model and field test shafts with measured τ_{sf} are collected for analysis to verify the proposed solution. The results indicate that the predicted τ_{sf} values are in good agreement with the measured τ_{sf} values. Extensive parametric studies are also conducted to investigate the effects of rock mass properties, shaft dimensions, and sidewall roughness parameters on the ultimate side shear resistance factor α_f (the ratio of τ_{sf} to the unconfined compressive strength of intact rock σ_c) of rock socketed shafts. The results indicate that the α_f increases with rock constant m_i , geological strength index (GSI) and asperity half chord length λ , but decreases with shaft diameter B , σ_c and disturbance D . Furthermore, the α_f first increases and then decreases with the asperity inclination angle δ , resulting in an optimum asperity inclination angle δ_{op} . Therefore, it is important to reduce the disturbance of rock mass from construction and select a proper asperity inclination angle when designing a rock socketed shaft.

1. Introduction

Rock socketed shafts are economical and efficient foundations to support various structures on different types of rock masses.^{1,2} The loads from structures are transferred through shaft sidewall and shaft end to the rock mass, leading to gradually reduced settlements along the depth direction. Due to the small settlement at the shaft end, the end resistance cannot reach its ultimate bearing capacity and is only partially mobilized, while the shear resistance from the sidewall dominates the total resistance at service loads.³ Therefore, the shear resistance from the shaft sidewall is of particular interest for the design of rock socketed shafts.

Numerous empirical methods have been proposed for determining the side shear resistance of rock socketed shafts based on laboratory and/or field tests. Most of those empirical methods relate the ultimate

side shear resistance τ_{sf} to the unconfined compressive strength of intact rock σ_c and can be generalized as

$$\tau_{sf} = m\sigma_c^k \quad (1)$$

where m and k are constants. Table 1 lists some of the empirical equations by various authors. As can be seen from Table 1, the constants m and k can change significantly between different empirical equations. This is because the empirical equations were developed based on specific databases. Therefore, although the empirical relations are useful for design practices, none of them can be applied to various locations. Also, the empirical relations do not effectively consider the effects of shaft dimensions and sidewall roughness on the ultimate side shear resistance.

As mentioned by many researchers,^{16–19} the shear behavior on a rough joint is accompanied by dilation across the interface, which leads to higher confining stress on the interface and thus the increase of the

* Corresponding author.

E-mail address: lyzhang@email.arizona.edu (L. Zhang).

<https://doi.org/10.1016/j.ijrmms.2022.105231>

Received 22 March 2022; Received in revised form 15 August 2022; Accepted 28 September 2022

Available online 7 October 2022

1365-1609/© 2022 Elsevier Ltd. All rights reserved.

Table 1
Empirical relationships for ultimate side shear resistance of rock socketed shafts.

Source	Equation
Kaderabek and Reynolds ⁴	$\tau_{sf} = 0.3\sigma_c$
Reese and O'Neill ⁵	$\tau_{sf} = 0.15\sigma_c$
Rosenberg and Journeaux ⁶	$\tau_{sf} = 0.34\sigma_c^{0.51}$
Horvath and Kenney ⁷	$\tau_{sf} = 0.21\sigma_c^{0.5}$
Meigh and Wolski ⁸	$\tau_{sf} = 0.22\sigma_c^{0.6}$
Williams ⁹	$\tau_{sf} = 0.44\sigma_c^{0.36}$
Horvath ¹⁰	$\tau_{sf} = (0.2 \sim 0.3)\sigma_c^{0.5}$
Rowe and Armitage ¹¹	$\tau_{sf} = 0.45\sigma_c^{0.5}$, for shafts with roughness R1, R2 or R3 $\tau_{sf} = 0.60\sigma_c^{0.5}$ for shafts with roughness R4
Kulhawy and Phoon ¹²	$\tau_{sf} = (0.22 \sim 0.67)\sigma_c^{0.5}$
Carrubba ¹³	$\tau_{sf} = (0.13 \sim 0.25)\sigma_c^{0.5}$
Zhang and Einstein ¹⁴	$\tau_{sf} = 0.2\sigma_c^{0.5}$, for smooth socketed shafts $\tau_{sf} = 0.8\sigma_c^{0.5}$ for rough socketed shafts
Xu et al. ¹⁵	$\tau_{sf} = 0.36\sigma_c^{0.39}$

shear strength. To better understand the side shear failure mechanism of rock socketed shafts, the stiffness of the rock mass normal to the joint or interface is often assumed constant in the laboratory, which is referred to as the constant normal stiffness condition.¹⁷ And various direct shear apparatuses based on the constant or variable normal stiffness conditions^{17,18} have been developed and applied to investigate the side shear behavior of rock socketed shafts. According to the laboratory tests using direct shear apparatuses with the constant normal stiffness design,¹⁷⁻¹⁹ the joint or interface roughness has a significant influence on the side shear behavior of rock socketed shafts.

To deal with shaft dimensions and sidewall roughness, numerical methods were adopted by different researchers to analyze shafts socketed in rock mass. Leong and Randolph²⁰ performed finite element analysis (FEA) of rock socketed shafts by using a model developed from principles of tribology to model the shaft-rock mass interface. Chung et al.²¹ used the ADINA FEA program to quantify the effects of spatial variability on the side shear resistance and end bearing capacity of rock-socketed shafts. It should be noted that the FEA models often use a friction angle to represent the behavior of the shaft-rock mass interface and cannot well capture the effect of the shaft sidewall roughness. Considering that, Gutiérrez-Ch et al.²² used a 3D discrete element method (DEM) model to analyze rock socketed shafts with different sidewall roughness. Note that a limited number of particles are often used in the 3D DEM modeling to ensure time efficiency,²² and the 3D DEM simulation might be time-consuming when analyzing a rock socketed shaft with small asperities and large dimensions.

On the other hand, theoretical studies were also performed to analyze the side shear resistance of rock socketed shafts. Considering the constant normal stiffness condition adopted in the laboratory tests, Johnston and Lam²³ simplified the rough sidewall as a series of continuous triangular concrete-rock joints and analyzed the shear behavior of one concrete-rock joint under four stages: (1) initial sliding, (2) subsequent sliding, (3) initial shearing, and (4) subsequent shearing.

Seidel and Haberfield²⁴ assumed the rough sidewall as triangular asperities and used the slope-stability solution of Sokolovskii^{25,26} to analyze the shearing mechanism of one representative triangular asperity. However, Seidel and Haberfield²⁴ used a planar solution based on the Mohr-Coulomb criterion and thus ignored the three-dimensional (3D) stress condition and the non-linear strength of rock masses. More recently, the approach of Seidel and Haberfield²⁴ was also extended by Zhao et al.,²⁷ Zhang et al.,²⁸ and Hou et al.²⁹ to study the shear failure mechanism and load-transfer curve of rock socketed shafts. Another interesting work was done by Serrano and Olalla³⁰ and it assumed a linear distribution for the vertical and horizontal stresses around a rock socketed shaft and determined the failure stress along the sidewall based on the Hoek-Brown (HB) criterion. Considering the increases in radial stress due to the expansion of the shaft-rock mass interface, Dai et al.³¹ modified the work of Serrano and Olalla.³⁰ It should be noted that the existing analytical solutions are based on a two-dimensional (2D) plane strain assumption and ignore the 3D strength of rock mass.

In this paper, an analytical solution is proposed to determine the ultimate side shear resistance of rock socketed shafts with consideration of the 3D strength of rock mass and shaft sidewall roughness. To do that, the profile of the shaft sidewall is simplified as triangular asperities and micro-mechanical analysis is performed to analyze the shearing of one representative asperity. As the rock mass around the shaft is under a 3D stress condition, the rock mass is assumed to be governed by a 3D version of the HB criterion and the equilibrium equations in a cylindrical coordinate system are utilized to derive the governing equations. The failure state of the triangular asperity is analyzed by the method of characteristics and the unit ultimate side shear resistance is further derived by considering the constant normal stiffness condition and the dilation (lateral displacement) of the shaft-rock mass interface due to sliding. To consider the widely used roughness factor (RF), a relation between RF and the parameters of equivalent triangular asperities is derived. Twenty-four (24) model and field test shafts are collected for analysis to verify the validity of the proposed solution. Finally, parametric studies based on the proposed solution are conducted to investigate the effects of rock mass properties, shaft dimensions, and sidewall roughness parameters on the ultimate side shear resistance of rock socketed shafts.

2. Basic assumptions

In this study, the following assumptions are made:

- 1) As often done by other studies,^{23,24,29} the sidewall of a rock socketed shaft is assumed to consist of triangular asperities along the shear direction as shown in Fig. 1. A triangular asperity i can be characterized by three geometric parameters: asperity inclination angle δ_i , half chord length λ_i and chord height h_i . And the relationship among those three parameters is^{23,24}

$$h_i = \lambda_i \tan \delta_i \quad (2)$$

Since h_i , λ_i and δ_i can change along the pile, the average values of

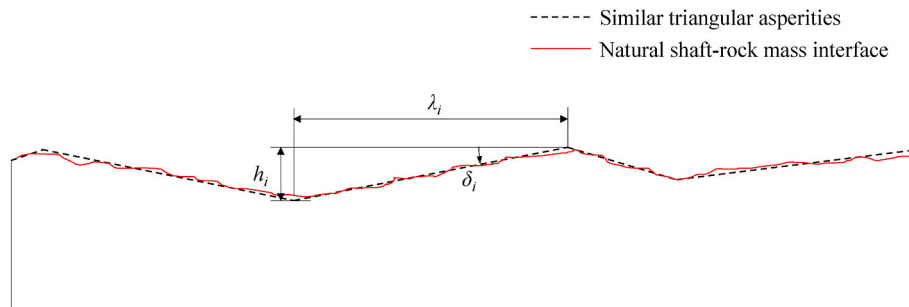


Fig. 1. Natural shaft-rock mass interface and triangular asperities.

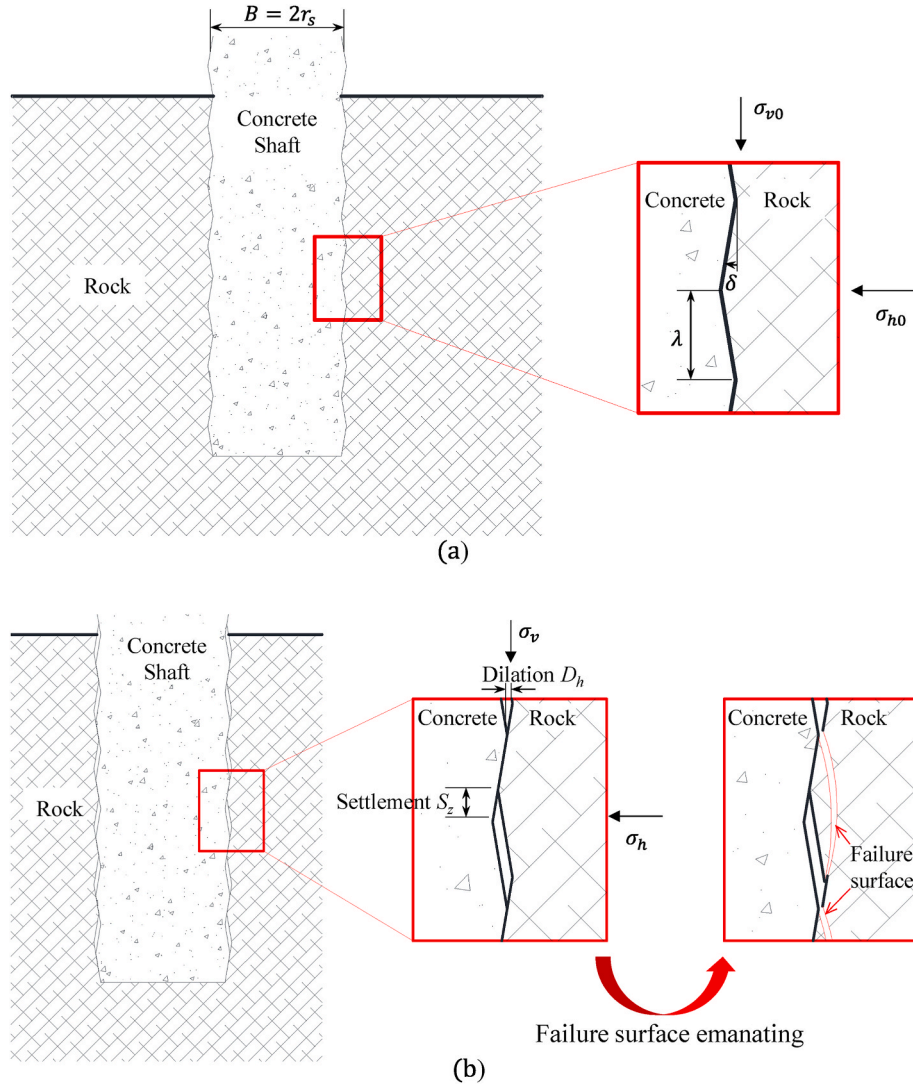


Fig. 2. Side shear behavior of a rock-socketed shaft: (a) before loading; and (b) failure after settlement.

them within a specific distance can be used for applying the proposed solution. Details on the determination of h , λ and δ will be discussed in section 4.2.

- 2) To be simple, only the asperities along the vertical (shear) direction are considered as shown in Fig. 2. And one representative asperity at a specific depth is analyzed to determine the τ_{sf} of the sidewall. The effect of the base displacement on the τ_{sf} of the sidewall is ignored as it has little influence on the failure stress state of the rock mass near the sidewall.
- 3) In this study, the joints of rock mass are assumed to be evenly distributed within the rock mass as shown in Fig. 2. Therefore, the rock mass is taken as an isotropic material governed by a 3D HB criterion, the newly modified GZZ criterion, whose expression is (Chen et al. ^{32,33})

$$f = \left[\left(\frac{q}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q}{\sigma_c} \right)^{1/a-1} \right] (I_1^* I_2^* - 9I_3^*) - 2m_b^2 I_3^* = 0 \quad (3)$$

where

$$q = \sqrt{I_1^{*2} - 3I_2^*} \quad (4a)$$

$$I_2^* = \sigma_1^* \sigma_2^* + \sigma_2^* \sigma_3^* + \sigma_3^* \sigma_1^* \quad (4b)$$

$$I_1^* = 3p^* = \sigma_1^* + \sigma_2^* + \sigma_3^* \quad (4c)$$

$$I_3^* = \sigma_1^* \sigma_2^* \sigma_3^* \quad (4d)$$

$$\sigma_i^* = \sigma_i + \frac{\sigma_c}{m_b} s, i = 1, 2, 3 \quad (4e)$$

$$m_b = \exp\left(\frac{GSI - 100}{28 - 14D}\right) m_i \quad (4f)$$

$$s = \exp\left(\frac{GSI - 100}{9 - 3D}\right) \quad (4g)$$

$$a = 0.5 + \frac{1}{6} \left(e^{-\frac{GSI}{15}} - e^{-\frac{20}{3}} \right) \quad (4h)$$

in which σ_1 , σ_2 and σ_3 = major, intermediate, and minor effective principal stresses, respectively; m_i = a constant for the intact rock; m_b , a and s = parameters of the rock mass; GSI = geology strength index; D = a disturbance factor representing the effect of construction on the rock mass.

- 4) Associate flow rule is adopted here to determine the velocity field, and thus the velocity field coincides with the stress field.

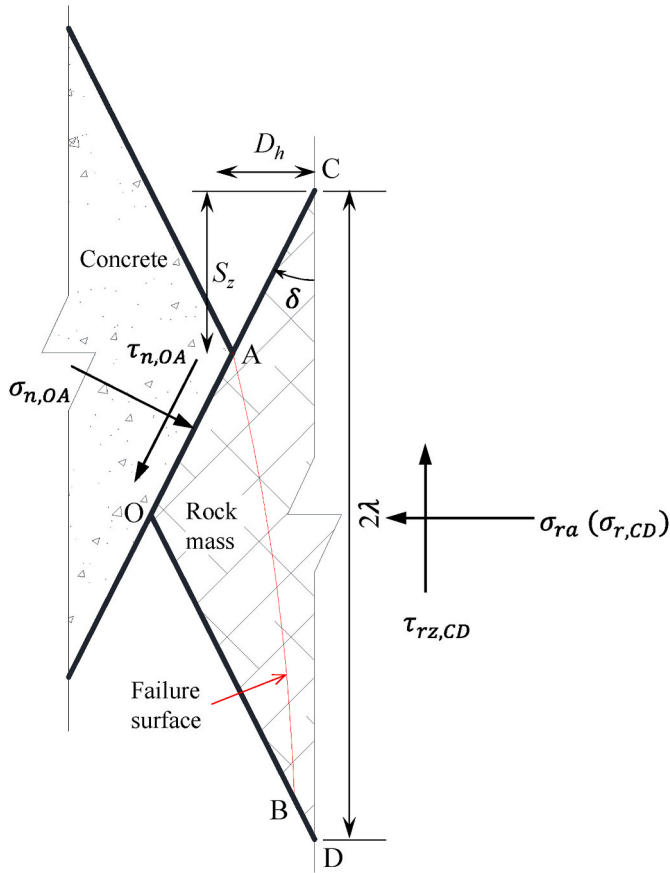


Fig. 3. Failure state of a triangular asperity.

- 5) The intermediate principal stress involved in the adopted failure criterion is determined by considering the elastic circumferential strain as³³

$$\sigma_\theta = E_{rm}\epsilon_\theta^e + \mu(\sigma_r + \sigma_z) + \sigma_{\theta 0} - \mu(\sigma_{r0} + \sigma_{z0}) \quad (5a)$$

$$\epsilon_\theta^e = -\frac{u_r}{r} \quad (5b)$$

where σ_r , σ_θ and σ_z = effective stresses along the radial, circumferential and vertical directions, respectively; E_{rm} and μ = Young's modulus and Poisson's ratio of the rock mass, respectively; σ_{r0} , $\sigma_{\theta 0}$ and σ_{z0} = initial values of σ_r , σ_θ and σ_z , respectively; ϵ_θ^e = elastic circumferential strain; u_r = radial displacement; and r = radial coordinate. The effect of σ_θ is to increase the bearing capacity. For convenience and conservative consideration, ϵ_θ^e is taken as zero and thus

$$\sigma_\theta = \mu(\sigma_r + \sigma_z) + \sigma_{\theta 0} - \mu(\sigma_{r0} + \sigma_{z0}) \quad (6)$$

- 6) The concrete is considered as a rigid material compared to the rock mass and the failure of the shaft-rock mass interface only takes place within the rock mass as shown in Fig. 2(b), this micro-scale mechanism is often adopted by similar research.^{27–29}

- 7) The friction angle φ of the shaft-rock mass interface is related to the stress level and rock mass properties and thus can be expressed as³⁴

$$\varphi = C_m \rho \quad (7)$$

where ρ = instant internal friction angle of the rock mass close to the shaft-rock mass interface; C_m = a constant related to shaft material and can be taken as 0.75 for concrete shafts.³⁴

- 8) At the failure state of the sidewall, relative sliding between the rock mass and the shaft occurs as shown in Fig. 2(b). Only part of the rock asperity contacts with the shaft and a failure surface forms within the triangular asperities below the contacting zone. The rock mass outside the asperities stays within the elastic state. The stress and deformation of the rock mass outside the asperities can be determined by using the constant normal stiffness condition,²⁴ as detailed in section 3.1.

3. Micromechanics-based model for ultimate side shear resistance of rock socketed shafts

The conditions of a rock socketed shaft with a diameter of B before and after loading are shown in Fig. 2. As shown in Fig. 2(a), the rock mass is under the vertical stress σ_{v0} and the horizontal stress σ_{h0} in the initial state before loading, and the rock mass and concrete shaft are in complete contact with no relative displacement between them. To be simple, the initial stress state before loading can be determined by

$$\sigma_{v0} = \gamma_s H_s + \gamma_r H_r \quad (8a)$$

$$\sigma_{h0} = k_0 \sigma_{v0} \quad (8b)$$

where γ_s and γ_r = unit weight of the soil and rock mass, respectively; H_s and H_r = thickness of the soil and rock mass layers, respectively; and k_0 = coefficient of lateral earth pressure at rest.

Once a vertical load is applied on the top of the shaft, a relative displacement S_z will take place between the shaft and the rock mass, and the rock mass would be pushed outward by a lateral displacement, i.e., dilation D_h . Due to the relative displacement, only the upper part of each triangular asperity remains in contact with the concrete shaft, and highly concentrated stress is expected to occur around the contact zone. Once the concentrated stress exceeds the failure stress of the rock mass, a failure surface will generate around the contact area and gradually emanate from the upper end to the lower end of the asperity. Once the failure surface penetrates through the rock mass, a large displacement is expected to take place and the side shear decreases due to the decrease of the contact zone, and thus the asperity fails (Fig. 2(b)). The following will develop a micromechanical model to analyze the failure state of the triangular asperities so as to determine the ultimate side shear resistance of a rock socketed shaft.

3.1. Constant normal stiffness condition

As discussed earlier, the shaft would push the rock mass outward during loading and the rock mass outside the asperities would remain elastic as the dilation D_h (radial displacement) is rather small as shown in Fig. 2. Therefore, a so-called constant normal stiffness condition proposed by Johnston and Lam²³ is utilized to determine the radial stress of the rock mass during loading. Specifically, the rock mass outside the asperity zone is taken as springs to hold the rock mass within the asperity and the global normal stiffness K of the surrounding rock mass (the assumed springs) is assumed as constant^{23,24} as the surrounding rock mass is outside the failure surface. In this regard, based on the cavity expansion theory,²³ the normal stiffness K of the assumed springs (surrounding rock mass outside the asperity) is constant. In particular, the K is related to the shaft radius r_s , the secant modulus and Poisson's ratio of rock mass and can be defined using the cylindrical cavity expansion theory as²³

$$K = \frac{E_{rm}}{(1 + \mu)r_s} = \frac{\Delta\sigma_{ra}}{\Delta r_s} \quad (9)$$

where $\Delta\sigma_{ra}$ = increment of radial stress outside the asperity; r_s = shaft radius; and Δr_s = change of r_s . E_{rm} can be estimated following Hoek and Diederichs³⁵

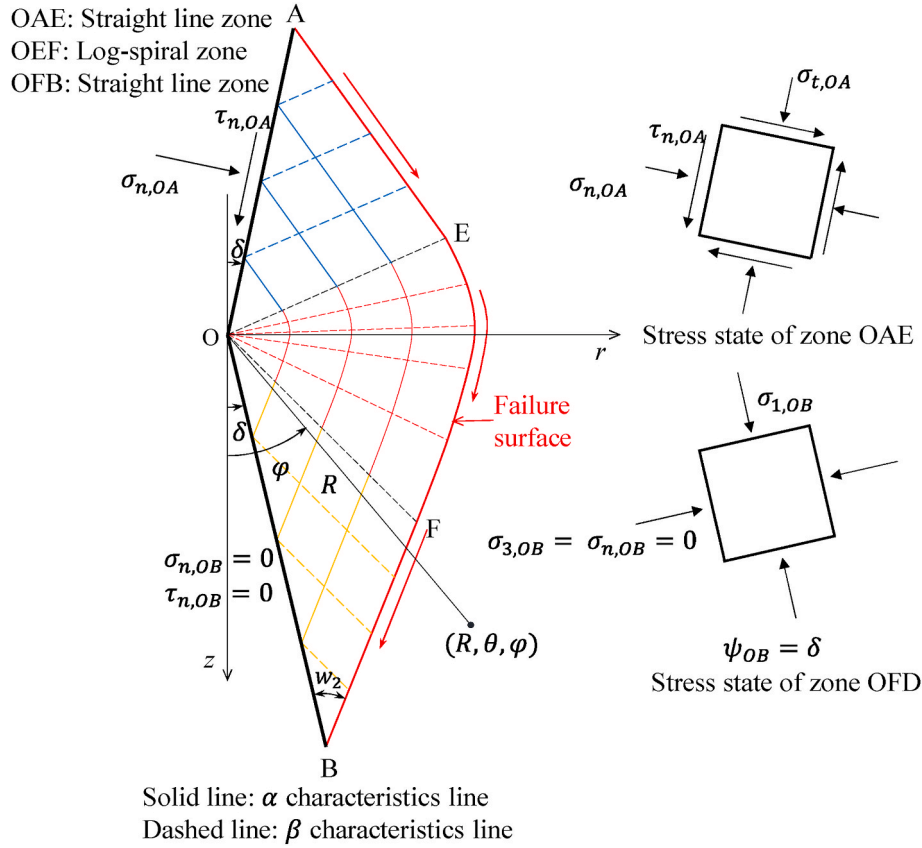


Fig. 4. Characteristics lines within asperity and stress states of constant-state zones.

$$E_m(\text{GPa}) = 100 \left[\frac{1 - D/2}{1 + e^{(75+25D-GSI)/11}} \right] \quad (10)$$

It should be noted that the deformation modulus, as the secant modulus, is used because the stress-strain relation of rock mass is non-linear even within the elastic stage.

Under this context, the radial stress in the rock mass outside the asperity after loading is

$$\sigma_{ra} = \sigma_{ra0} + KD_h = \sigma_{ra0} + KS_z \tan \delta \quad (11)$$

where σ_{ra0} = initial radial stress outside the asperity before loading and $\sigma_{ra0} = \sigma_{h0}$; and D_h = dilation (radial displacement) of the shaft-rock mass interface.

3.2. Micro-mechanical analysis of one triangular asperity

Fig. 3 shows the failure state of a triangular asperity, with the left side being the concrete shaft and the right side being the rock mass. Before loading, the rock mass and concrete shaft are compactly contacted with point A coinciding with point C. Due to a relative settlement, the rock mass is radially pushed out by a dilation (radial displacement) of D_h and only the OA zone has the concrete shaft and rock mass in contact. Due to the S_z and D_h , both the normal and shear stresses on CD, $\sigma_{r,CD}$ and $\tau_{r,CD}$, would be enhanced. The red curve within the rock mass is the potential failure surface. Note that it might take some relative settlement to form the failure surface that penetrates the asperity. But once the failure surface is formed completely, the stress state within OAB could be determined by the boundary conditions on OA, OB and the failure surface AB and the governing equation of OAB, which will be determined by using the method of characteristics and discussed in section 3.3. After the formation of the failure surface, the failure zone OAB would continue sliding along AB due to the load from the shaft and

thus the $\tau_{r,CD}$ would decrease due to the decrease of the contact area AB. In this regard, the side shear stress of the asperity when the failure surface just penetrates the asperity can be regarded as the ultimate side shear resistance, which is in agreement with other similar research.^{28,29}

Using the force equilibrium equations in the vertical and horizontal directions

$$\sigma_{n,OA} \frac{\lambda - S_z}{\cos \delta} \cos \delta - \tau_{n,OA} \frac{\lambda - S_z}{\cos \delta} \sin \delta - 2\lambda \sigma_{r,CD} = 0 \quad (12a)$$

$$-\sigma_{n,OA} \frac{\lambda - S_z}{\cos \delta} \sin \delta - \tau_{n,OA} \frac{\lambda - S_z}{\cos \delta} \cos \delta + 2\lambda \tau_{r,CD} = 0 \quad (12b)$$

we have

$$\sigma_{r,CD} = \frac{\lambda - S_z}{2\lambda} (\sigma_{n,OA} - \tau_{n,OA} \tan \delta) \quad (13a)$$

$$\tau_{r,CD} = \frac{\lambda - S_z}{2\lambda} (\sigma_{n,OA} \tan \delta + \tau_{n,OA}) \quad (13b)$$

with $\sigma_{r,CD}$ and $\tau_{r,CD}$ expressed as a function of S_z , $\sigma_{n,OA}$ and $\tau_{n,OA}$, the next step is to determine S_z , $\sigma_{n,OA}$ and $\tau_{n,OA}$ when the failure surface just penetrates the asperity.

3.3. Determination of failure stresses on contacting zone OA

3.3.1. Governing equations using method of characteristics

As shown in Fig. 3, for asperity COD, there is no pressure on AC and OD after radial displacement D_h . According to similar research,²⁷⁻²⁹ the stress state of AOB within a rough asperity can be determined by utilizing the method of characteristics.^{25,26} Fig. 4 shows the characteristics lines of a triangular asperity under the failure condition; the solid lines are the α characteristics lines and the dashed lines denote the β

characteristics lines. Typically, a characteristics-line field ignoring the gravity can be decomposed into two constant-state regions (straight line zones OAE and OFB) and one logarithmic spiral region (OEF). The external loads (normal stress $\sigma_{n,OA}$ and shear stress $\tau_{n,OA}$) are applied on the OA side of the obtuse triangular wedge. And the failure surface is curve AEFB consisting of two straight lines AE and FB and one log-spiral curve EF. Since OAE and OFB are two constant-state regions, the key is to solve the stresses within OEF. Assuming that the r and z axes are in the horizontal and vertical directions with the vertex O as the origin and ignoring the weight of the rock mass, the equilibrium equations in the cylindrical coordinate system can be expressed as

$$\frac{\partial \sigma_r}{\partial r} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} = 0 \quad (14a)$$

$$\frac{\partial \tau_{rz}}{\partial r} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{rz}}{r} = 0 \quad (14b)$$

where τ_{rz} is the shear stress within the rz -plane.

The differential equation of the α characteristics line in a clockwise rotation motion is

$$\frac{dr}{dz} = \tan(\psi - \omega) \quad (15)$$

where ψ denotes the inclination of the major effective principal stress σ_1 relative to the positive axis z ; and ω denotes the angle between the direction of the major effective principal stress and the two characteristic lines and can be expressed as

$$\omega = \frac{\pi}{4} - \frac{\rho}{2} \quad (16)$$

The three stresses within the rz plane can be obtained by considering stress rotation of an angle ψ following Chen et al.³⁶ as

$$\sigma_r = p_L - q_L \cos 2\psi \quad (17a)$$

$$\sigma_z = p_L + q_L \cos 2\psi \quad (17b)$$

$$\tau_{rz} = q_L \sin 2\psi \quad (17c)$$

where the two Lambe parameters p_L and q_L are

$$p_L = (\sigma_1 + \sigma_3)/2 \quad (18a)$$

$$q_L = (\sigma_1 - \sigma_3)/2 \quad (18b)$$

Combining Eqs. (14), (17) and (18), the following governing equation along the α characteristic line can be obtained as

$$\cos \rho dp_L - 2q_L d\psi + \frac{(p_L - \sigma_\theta) \cos \rho - q_L \cos(2\psi + \rho)}{2r} dr + \frac{q_L \sin(2\psi + \rho) - (p_L - \sigma_\theta) \sin \rho}{2r} dz = 0 \quad (19)$$

In the polar coordinate system (R, θ, φ) in Fig. 4, the coordinates r , z and the direction of the major effective principal stress ψ of the α characteristic line can be expressed as

$$r = R \sin \varphi \quad (20a)$$

$$z = R \cos \varphi \quad (20b)$$

$$\psi = \varphi - \omega = \varphi + \frac{\rho}{2} - 45^\circ \quad (20c)$$

Performing differentiation to Eq. (20) and considering Eq. (15), one has

$$dr = \sin \varphi dR + R \cos \varphi d\varphi \quad (21a)$$

$$dz = \cos \varphi dR - R \sin \varphi d\varphi \quad (21b)$$

$$\frac{dR}{d\varphi} = -R \tan \rho \quad (21c)$$

Combining Eqs. (19) and (21), one has

$$\cos \rho dp_L - 2q_L d\psi + C_\varphi d\varphi = 0 \quad (22)$$

where

$$C_\varphi = \frac{(p_L - \sigma_\theta) \cos \rho - q_L \cos(2\psi + \rho)}{2 \sin \varphi} (\sin \varphi \tan \rho + \cos \varphi) + \frac{q_L \sin(2\psi + \rho) - (p_L - \sigma_\theta) \sin \rho}{2 \sin \varphi} (\cos \varphi \tan \rho - \sin \varphi) \quad (23)$$

It should be pointed out that each point on the α characteristic line should satisfy the newly modified GZZ criterion (Eq. (3)) and Eq. (6). Therefore

$$df = \frac{\partial f}{\partial \sigma_1^*} d\sigma_1 + \frac{\partial f}{\partial \sigma_2^*} d\sigma_2 + \frac{\partial f}{\partial \sigma_3^*} d\sigma_3 = 0 \quad (24a)$$

$$d\sigma_\theta = d\sigma_2 = \mu(d\sigma_r + d\sigma_z) = \mu(d\sigma_1 + d\sigma_3) = 2\mu dp_L \quad (24b)$$

Substituting Eq. (24b) into Eq. (24a) yields

$$d\sigma_1 = \frac{2}{1 + C_{13}} dp_L \quad (25a)$$

$$d\sigma_2 = \frac{2C_{12}}{1 + C_{13}} dp_L = 2\mu dp_L \quad (25b)$$

$$d\sigma_3 = \frac{2C_{13}}{1 + C_{13}} dp_L \quad (25c)$$

where

$$C_{12} = -\mu \frac{\frac{\partial f}{\partial \sigma_1^*} - \frac{\partial f}{\partial \sigma_3^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}} \quad (26a)$$

$$C_{13} = -\frac{\frac{\partial f}{\partial \sigma_1^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}} \quad (26b)$$

The details of Eq. (26) are provided in Appendix A1.

Therefore, the three principal stresses can be determined if p_L is known. In this regard, the stress state of the axisymmetric problem solely depends on p_L . And the instant friction angle ρ can be expressed by the stresses under axisymmetric condition following Chen et al.³⁶ as

$$\sin \rho = \frac{dq_L}{dp_L} = \frac{d\sigma_1 - d\sigma_3}{d\sigma_1 + d\sigma_3} = \frac{1 - C_{13}}{C_{13} + 1} \quad (27)$$

Since φ is a function of ψ and ρ as indicated in Eq. (20c) and ρ can be determined from p_L based on Eq. (27), Eq. (22) can be further simplified as

$$\cos \rho dp_L - 2q_L d\psi + C_\varphi d\psi - \frac{C_\varphi}{2} d\rho = \left(\cos \rho + \frac{C_\varphi}{2} \frac{\partial \rho}{\partial p_L} \right) dp_L - (2q_L + C_\varphi) d\psi = 0 \quad (28a)$$

Reorganizing Eq. (28a) yields

$$\frac{dp_L}{d\psi} = \frac{2q_L + C_\varphi}{\cos \rho + \frac{C_\varphi}{2} \frac{\partial \rho}{\partial p_L}} = \frac{\sigma_1 - \sigma_3 + C_\varphi}{\cos \rho + \frac{C_\varphi}{2} \frac{\partial \rho}{\partial p_L}} \quad (28b)$$

Substituting Eq. (28b) into Eq. (25), the three principal stresses can be determined

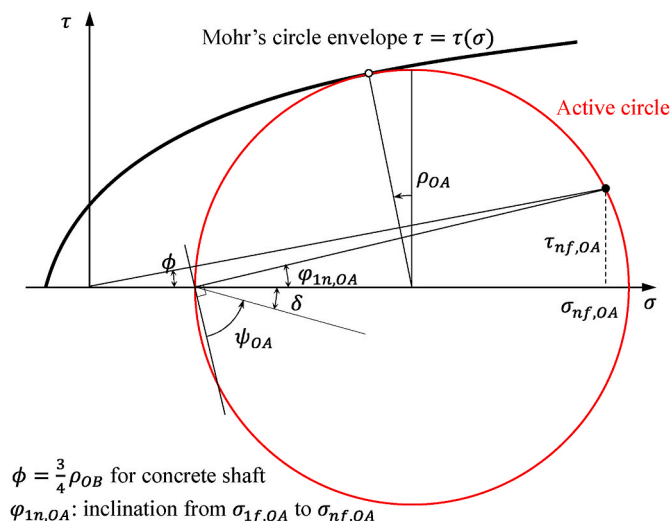


Fig. 5. Active Mohr circle for region OAE.

$$\frac{d\sigma_1}{d\psi} = \frac{2}{1 + C_{13}} \frac{(\sigma_1 - \sigma_3 + C_\varphi)}{\cos \rho + \frac{C_\varphi}{2} \frac{\partial \rho}{\partial p_t}} \quad (29a)$$

$$\frac{d\sigma_2}{d\psi} = \frac{2C_{12}}{1+C_{13}} \frac{(\sigma_1 - \sigma_3 + C_\varphi)}{\cos \rho + \frac{C_\varphi}{\gamma} \frac{\partial \rho}{\partial n}} \quad (29b)$$

$$\frac{d\sigma_3}{d\psi} = \frac{2C_{13}}{1+C_{13}} \frac{(\sigma_1 - \sigma_3 + C_\varphi)}{\cos \rho + \frac{C_\varphi}{\gamma} \frac{\partial \rho}{\partial \psi}} \quad (29c)$$

Performing differentiation to Eq. (27) yields

$$\frac{d\rho}{dp_L} = -\frac{2}{\cos \rho (C_{13} + 1)^2} \frac{dC_{13}}{dp_L} \quad (30a)$$

$$\frac{dC_{13}}{dp_L} = \frac{2}{1+C_{13}} \frac{dC_{13}}{d\sigma_1} + \frac{2C_{12}}{1+C_{13}} \frac{dC_{13}}{d\sigma_2} + \frac{2C_{13}}{1+C_{13}} \frac{dC_{13}}{d\sigma_3} \quad (30b)$$

The details of Eq. (30) are given in Appendix A2.

Substituting Eq. (28b) into Eq. (30), one has

$$\frac{d\rho}{d\psi} = \frac{\partial\rho}{\partial p_L} \frac{dp_L}{d\psi} = \frac{(\sigma_1 - \sigma_3 + C_\varphi)}{\cos\rho + \frac{C_\varphi}{2} \frac{\partial\rho}{\partial p_I}} \frac{\partial\rho}{\partial p_L} \quad (31a)$$

$$\frac{d\varphi}{d\psi} = 1 - \frac{1}{2} \frac{d\rho}{d\psi} = 1 + \frac{(\sigma_1 - \sigma_3 + C_\varphi)}{2 \cos \rho + C_\varphi \frac{\partial \rho}{\partial p_L}} \frac{\partial \rho}{\partial p_L} \quad (31b)$$

It should be noted that Eqs. (29) and (31) are the governing equations along the α characteristic line (failure surface) within BOC with ψ as the basic variable. To determine the stresses and the shape of the α characteristics line, a Runge-Kutta-based approach is proposed to numerically integrate Eqs. (29) and (31) from boundary OF to boundary OE as shown in Fig. 4. Therefore, the next step is to determine the stresses, ρ , ω and ψ on boundary OF and the ψ on boundary OE.

3.3.2. Determination of boundary conditions

3.3.2.1. Boundary OF. Since OFB is a constant-state zone, the stresses, instant friction angle ρ and direction of the major effective principal stress ψ should remain the same within it. As shown in Fig. 4, the normal and shear stresses on OB are zero. So, the direction of σ_1 on OF and OB should be along the direction of boundary OB, i.e.

$$\psi_{OF} = \psi_{OB} = \delta \quad (32)$$

As for the principal stresses on OF and OB, substituting $\sigma_3 = 0$ into Eqs. (3) and (6), one can obtain σ_1 and σ_2 . Once the stresses are determined, the instant friction angle ρ and φ_{OB} on OB and OF can be obtained with Eq. (27) and Eq. (20c), respectively.

3.3.3. Boundary OE

Similarly, OAE is a constant-state region and one can conveniently take the mechanical state (stresses, ρ and ψ) of it as that of OE. Once the failure surface is completely formed, a clockwise rotational movement around point O, i.e., the velocity field, is generated. As for the points within OAE, the active Mohr's circle can be plotted tangent to the strength curve based on the active moving mode as shown in Fig. 5.

At the failure state, the shear stress $\tau_{nf,OA}$ and normal stress $\sigma_{nf,OA}$ on OA can be related to the friction angle φ of the shaft-rock mass interface as

$$\tau_{nf,OA} = \sigma_{nf,OA} \tan \varphi \quad (33)$$

As shown in Fig. 5, the angle $\varphi_{1n,OA}$ between the major effective principal stress on OA $\sigma_{1f,OA}$ and the normal effective stress on OA $\sigma_{n,OA}$ (counterclockwise from $\sigma_{1f,OA}$ to $\sigma_{nf,OA}$) can be related to the stress state on OA as

$$\tau_{nf,OA} = \frac{\sigma_{1f,OA} - \sigma_{3f,OA}}{2} \sin 2\varphi_{1n,OA} = q_{Lf,OA} \sin 2\varphi_{1n,OA} \quad (34a)$$

$$\sigma_{\text{hf},OA} = \frac{\sigma_{\text{lf},OA} + \sigma_{\text{3f},OA}}{2} + \frac{\sigma_{\text{lf},OA} - \sigma_{\text{3f},OA}}{2} \cos 2\varphi_{1n,OA} = p_{\text{Lf},OA} + q_{\text{Lf},OA} \cos 2\varphi_{1n,OA} \quad (34b)$$

where $\sigma_{1f,OA}$ and $\sigma_{3f,OA}$ = major and minor effective principal stresses on OA at failure, respectively.

Combining Eqs. (33) and (34), one has

$$\varphi_{\text{ln},OA} = \frac{1}{2} \left[\sin^{-1} \left(\frac{p_{L_f,OA} \sin \varphi}{q_{L_f,OA}} \right) + \varphi \right] \quad (35)$$

And the angle from positive z -direction to OA is $180^\circ - \delta$. So, the inclination of $\sigma_{nf,OA}$ from positive z -direction is $90^\circ - \delta$ (counterclockwise). Therefore, the inclination from positive z -direction to $\sigma_{lf,OA}$, as shown in Fig. 5, is

$$\psi_{OA} = \psi_{OF} = 90^\circ - \delta - \varphi_{|n, OA} \quad (36)$$

It should be noted that ψ_{OA} can be determined by an iterative approach which will be discussed in section 3.4. As for the stresses, ρ and ψ within OAE, they should be determined by the integration of Eqs. (29) and (31).

3.4. Determination of ultimate side shear resistance

Once the failure stresses $\sigma_{nf,OA}$ and $\tau_{nf,OA}$ on OA are determined from section 3.3, substituting $\sigma_{nf,OA}$ and $\tau_{nf,OA}$ into Eqs. (11) and (13), one has the relative settlement at failure as

$$S_{\mathcal{F}} = \lambda \left[1 - 2 \frac{\lambda K \tan \delta + \sigma_{ra0}}{2\lambda K \tan \delta + \sigma_{nf, OA} (1 - \tan \delta \tan \varphi_{OA})} \right] \quad (37a)$$

$$\varphi_{\partial A} = C_m \rho_{\partial A} \quad (37b)$$

Substituting Eq. (37) back into Eq. (13) yields

$$\sigma_{rf,CD} = \frac{\lambda - S_{zf}}{2\lambda} \sigma_{nf,OA} (1 - \tan \delta \tan \varphi_{OA}) \quad (38a)$$

$$\tau_{sf} = \tau_{rf,CD} = \frac{\lambda - S_{zf}}{2\lambda} \sigma_{nf,OA} (\tan \delta + \tan \varphi_{OA}) \quad (38b)$$

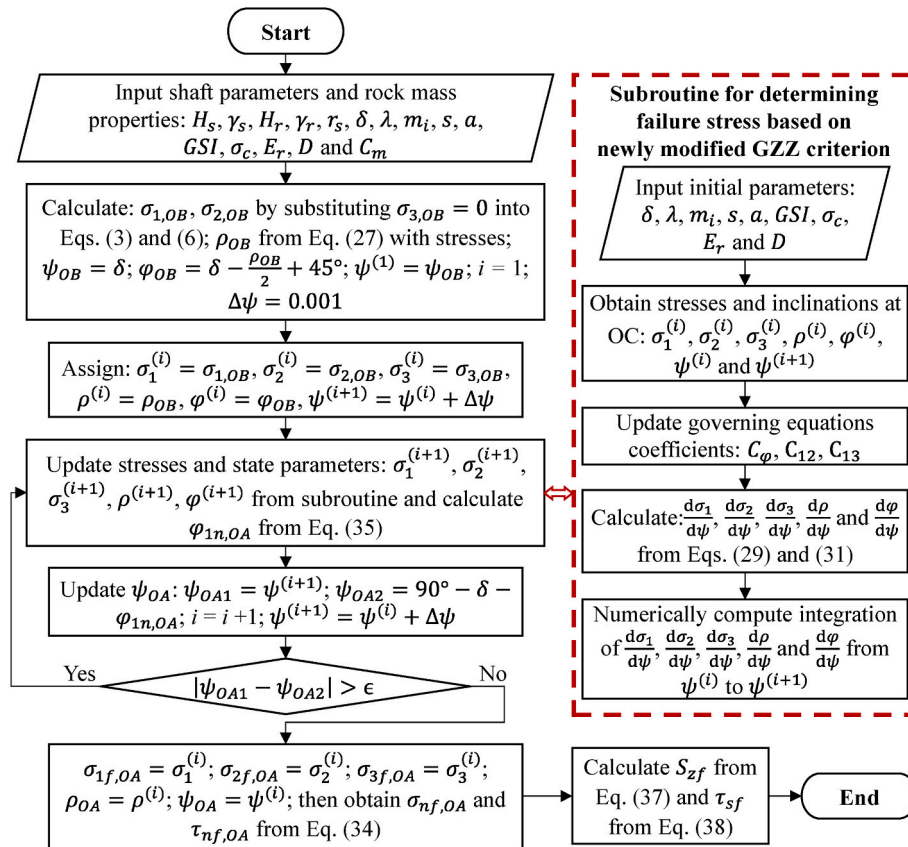


Fig. 6. Iterative solving procedure for proposed analytical solution.

Table 2

Summary of shaft tests and predictions from proposed solution.

#	Rock type	m_i	σ_c (MPa)	E_{rm} (MPa)	GSI	H_r (m)	B (mm)	λ (mm)	RF	h (mm)	δ (°)	τ_{sf} (MPa)				
												Measured	This study			Xu et al. ¹⁵
													$D = 0$	$D = 0.5$	$D = 1$	
1	Gypsum	8	6	608	85	0.5	50	20	0.000	0	1.00	1.140	1.046	0.970	0.812	0.724
2	Gypsum	8	6	608	85	0.5	50	20	0.040	1	2.86	2.234	1.957	1.733	1.336	0.724
3	Gypsum	8	6	608	85	0.5	50	20	0.082	2	5.82	2.822	2.682	2.313	1.720	0.724
4	Gypsum	8	6	608	85	0.5	50	20	0.127	3	8.91	2.689	3.042	2.597	1.910	0.724
5	Gypsum	8	6	608	85	0.5	50	20	0.232	5	15.60	2.932	3.145	2.660	1.963	0.724
6	Sandstone	17	6	*	50	1.0**	75	125	–	2.5	1.15	0.820	1.546	0.832	0.246	0.724
7	Sandstone	17	6	*	50	1.0**	160	125	–	2.5	1.15	0.940	1.528	0.812	0.238	0.724
8	Sandstone	17	6	*	50	1.0**	160	125	–	2.5	1.15	1.160	1.528	0.812	0.238	0.724
9	Sandstone	17	6	*	50	1.0**	315	125	–	2.5	1.15	0.890	1.497	0.779	0.225	0.724
10	Sandstone	17	6	*	50	1.0**	255	125	–	2.5	1.15	1.650	1.509	0.791	0.230	0.724
11	Sandstone	17	6	*	50	1.0**	160	125	–	2.5	1.15	1.130	1.528	0.812	0.238	0.724
12	Sandstone	17	6	*	50	1.0**	290	125	–	2.5	1.15	0.680	1.502	0.784	0.227	0.724
13	Sandstone	17	6	*	50	1.0**	210	125	–	7	3.21	1.120	1.652	0.890	0.264	0.724
14	Sandstone	17	6	*	50	1.0**	210	125	–	7	3.21	0.890	1.652	0.890	0.264	0.724
15	Sandstone	17	6	*	50	1.0**	210	125	–	7	3.21	1.200	1.652	0.890	0.264	0.724
16	Sandstone	17	6	*	50	1.0**	210	125	–	7	3.21	1.170	1.652	0.890	0.264	0.724
17	Sandstone	17	6	*	50	1.0**	315	125	–	10	4.57	1.410	1.698	0.915	0.273	0.724
18	Sandstone	17	6	*	50	1.0**	710	125	–	10	4.57	0.650	1.679	0.895	0.264	0.724
19	Mudstone	6	11.1	1170	70	1.0**	710	87.4	0.025	24.1	5.77	1.110	2.032	1.528	0.843	0.920
20	Mudstone	6	5.4	1050	70	1.0**	710	113.0	0.036	12.7	6.41	1.110	1.175	0.849	0.446	0.695
21	Mudstone	6	10.4	741	70	1.0**	710	35.5	0.039	13.0	20.12	1.090	1.855	1.351	0.786	0.897
22	Clay shale	6	1.2	134	73	9.2	760	70	–	3.8	3.11	0.106	0.178	0.148	0.096	0.387
23	Clay shale	6	3.1	212	66	7.2	760	70	–	6.0	4.90	0.430	0.410	0.316	0.176	0.560
24	Limestone	10	10.2	5500	74	4.0	760	70	–	3.1	2.54	1.553	2.968	2.341	1.405	0.891

*: E_{rm} is determined by using Eq. (10) and is thus related to the D value used; **: H_r is not available from the source and taken as 1 m.

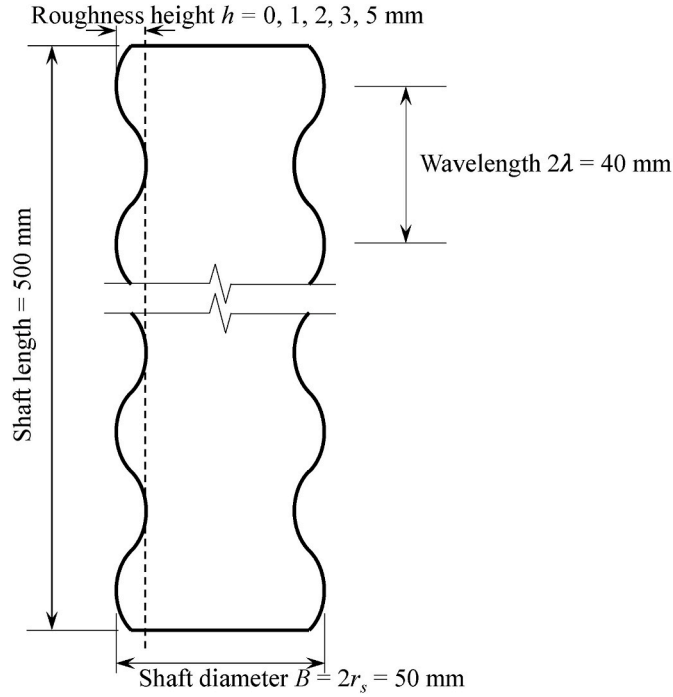


Fig. 7. Dimensions of model shafts by Dai et al. (2017).

Eq. (38b) gives the ultimate side shear resistance τ_{sf} of the rock socketed shaft. The solving procedure for the proposed solution is shown in Fig. 6.

4. Roughness parameters

4.1. Relation between roughness factor and parameters of equivalent triangular asperities

In practice, roughness factor (RF) is commonly used to describe the roughness of the sidewall of a rock-socketed shaft. Various methods have been proposed for quantifying the roughness of shaft sidewall (such as Myers³⁷; Tse and Cruden³⁸; Horvath et al.³⁹; Pells et al.⁴⁰). Among them, the RF defined by Horvath et al.³⁹ is the most widely used:

$$RF = \frac{\bar{h}}{\bar{r}_s} \frac{L_t}{L} \quad (39)$$

where $\bar{h}$ = average asperity height; L_t = total travel length along the sidewall profile; $\bar{r}_s$ = nominal shaft radius (average shaft radius); and L = nominal length of the socket.

To apply the proposed solution, it is important to relate the parameters λ , h and δ which are related to each other by Eq. (2) to the RF in a general way. Considering the equivalent triangular asperities with parameters h , λ and δ along the sidewall of a rock-socketed shaft, Eq. (39) can be simplified as

$$RF = \frac{h}{\bar{r}_s \cos \delta} = \frac{\lambda \tan \delta}{\bar{r}_s \cos \delta} \quad (40)$$

Therefore

$$\delta = \sin^{-1} \left(\frac{\sqrt{\lambda^2 + 4RF^2 \bar{r}_s^2} - \lambda}{2RF \times \bar{r}_s} \right), \text{ if } \lambda \text{ is known} \quad (41a)$$

$$\delta = \cos^{-1} \left(\frac{h}{RF \times \bar{r}_s} \right), \text{ if } h \text{ is known} \quad (41b)$$

Once the equivalent parameters h , λ and δ are determined from the RF based on Eq. (41), the proposed solution can be used by following the

procedure in Fig. 6 to determine the ultimate side shear resistance of a rock-socketed shaft.

4.2. Determination of roughness parameters

In engineering practices, the roughness parameters should be determined from the shaft-rock mass interface profile before using the proposed approach. To do that, the shaft-rock mass interface (or shaft sidewall) profile should be obtained from the site by using laser sensor⁴¹ or other methods. Details about the available laser sensor-based method or other methods for obtaining the shaft sidewall profile can be found in Park et al.⁴¹ Then, as shown in Fig. 1, a triangular asperity profile can be drawn to fit the obtained shaft sidewall profile and the h_i , λ_i and δ_i of each asperity i can be easily determined. To be simple, the average values of h_i , λ_i and δ_i of the shaft within the whole length of the rock-socketed shaft can be used as the h , δ and λ of the representative asperity. To consider the mutual effects among different asperities with different sizes, one can refer to Seidel and Haberfield²⁴ for a more refined method.

5. Validation of proposed solution

In this section, twenty-four (24) model and field test rock-socketed shafts with information on sidewall roughness and measured ultimate side shear resistance are used to validate the proposed solution. For each shaft, the shaft dimensions, rock mass properties and test results are collected and summarized in Table 2. According to AASHTO,⁴² if blasting is used for shaft construction, a disturbance factor close to 1.0 should be considered. If rock coring is used to drill a hole, much less disturbance is expected and a disturbance factor of 0 may be considered. If a down-hole hammer is used to construct the shaft, the disturbance is most likely between these two extremes. Since no information is available regarding the construction method, the disturbance factor (D) values of 0, 0.5, and 1.0 representing respectively the no disturbance, medium disturbance and severe disturbance of the rock mass from construction are considered. To highlight the importance of the roughness, the empirical equation of Xu et al.¹⁵ is also utilized to analyze the shafts.

5.1. Model shaft tests by Dai et al.⁴³

Dai et al.⁴³ conducted several model shaft tests in artificial rock to investigate the effect of sidewall roughness. The artificial rock was made using a mixture of sand, cement, gypsum powder and water, and its unconfined compressive strength, modulus and unit weight from the lab tests were 6 MPa, 608 MPa and 22 kN/m³, respectively. The model shafts were made of solid plexiglass, which has much higher strength and modulus than the rock, with dimensions shown in Fig. 7. All the shafts were 500 mm long and had a circular cross-section with a maximum diameter of 50 mm. Sinusoidal asperities with different heights (0, 1, 2, 3 and 5 mm) were grinded out on the sidewall to represent different degrees of roughness and the wavelength of the asperities was 40 mm for all shafts. Five shafts (#1 to #5 in Table 2) with five different roughness heights were tested and the results are summarized in Table 2.

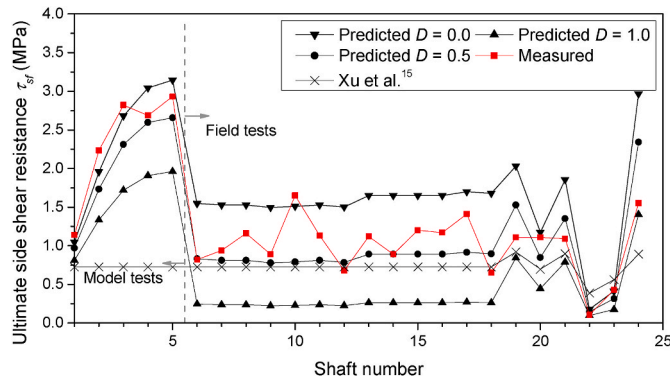
To make predictions on those shafts, $m_i = 8$ was used by taking the artificial rock as gypsum according to Hoek and Brown.⁴⁴ No information is available about the fracture condition of the artificial rock, but it is rational to assume that the rock is in a good condition. Therefore, a GSI = 85 is assigned to the rock. Since the shape of asperities is not triangular, the asperity inclination angle δ of each shaft is determined by using Eq. (41a). As for the zero-roughness shaft, a small asperity inclination angle $\delta = 1^\circ$ is assumed considering that the sidewall is not perfectly smooth.

Table 3Summary of shaft information from Williams and Pells.⁴⁵

#	Rock type	Shaft	m_i	σ_c (MPa)	E_r (MPa)	GSI	H_r (m)	B (mm)	Roughness class	Measured τ_{sf} (MPa)
6–12	Sandstone	A1, B1, B2, B3, B4, B5, E3	17	6	–	50	1.0 ^a	75–315	R1	0.680–1.650
13–16	Sandstone	A2, A4, C4, C5	17	6	–	50	1.0 ^a	210–210	R2	0.890–1.200
17–18	Sandstone	A3, D2	17	6	–	50	1.0 ^a	315–710	R3	0.650–1.410

^a H_r is not available from the source and taken as 1 m.**Table 4**Shaft roughness classifications (after Pells et al.⁴⁰).

Roughness class	Description
R1	Straight, smooth-sided socket; grooves or indentations less than 1 mm deep
R2	Grooves 1–4 mm deep, >2 mm wide, spacing 50–200 mm
R3	Grooves 4–10 mm deep, >5 mm wide, spacing 50–200 mm
R4	Grooves or undulations >10 mm deep, >10 mm wide, spacing 50–200 mm

**Fig. 8.** Comparisons of measured and predicted τ_{sf} values.**Table 5**

Rock mass properties, shaft dimensions, and roughness parameters for parametric analyses.

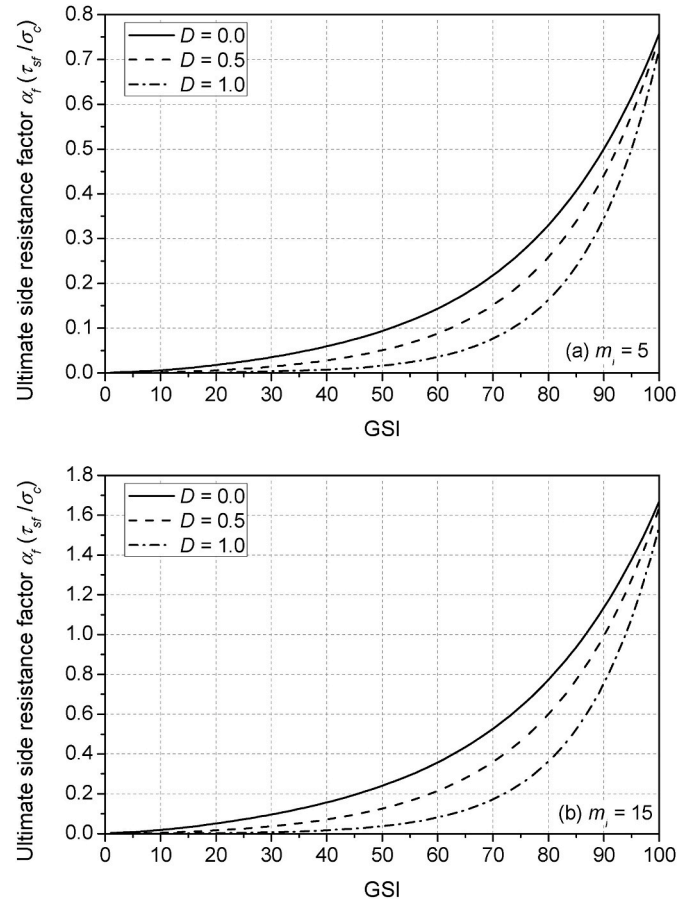
#	m_i	σ_c (MPa)	GSI	H_r (m)	B (m)	λ (mm)	δ (°)
a	5	5	75	2	1	20	10
b	15	5	75	2	1	20	10

5.2. Field tests by Williams and Pells⁴⁵

Williams and Pells⁴⁵ reported field tests on several shafts socketed in sandstone. Based on the degree of roughness, those shafts were classified into three categories as shown in Table 3. The degree of roughness is defined following Pells et al.⁴⁰ as shown in Table 4. The m_i of sandstone is taken as 17 according to Hoek and Brown,⁴⁴ and the σ_c from tests was 6 MPa. The modulus of the rock mass was estimated from Eq. (10) by assuming an average GSI of 50 for all the cases. The B is from Williams and Pells⁴⁵ and the H_r is taken as 1 m for each shaft. The average groove depth and spacing from Table 4 are taken as the λ and h for the shafts with different roughness degrees. Since no information about the RF is available, δ is determined from Eq. (2). The detailed information and test results of those shafts are also summarized in Table 2.

5.3. Full-size shaft tests by Horvath et al.⁴⁶

Three full-size shafts socketed into weak rock mass were investigated by Horvath et al.⁴⁶ to find proper methods for improving the

**Fig. 9.** Ultimate side shear resistance factor α_f versus GSI: (a) $m_i = 5$; and (b) $m_i = 15$

performance of rock socketed shafts. The σ_c and E_{rm} of the mudstone around shafts #19, #20 and #21 and the shaft dimensions are summarized in Table 2. The m_i of mudstone is determined following Hoek and Brown⁴⁴ and the GSI is estimated from the measured RQD using the following relation by Hoek et al.⁴⁷

$$GSI = 1.5J\text{Cond}_{89} + RQD/2 \quad (42)$$

in which, $J\text{Cond}_{89}$ is the joint condition rating defined by Bieniawski.⁴⁸ Since $J\text{Cond}_{89}$ was not measured, and an average value of 20 was assumed. Using the measured RQD value of 75 for all the cases (shafts #19, #20 and #21), the GSI was determined as 70 as shown in Table 2. Based on the average roughness height h and RF values, the δ and λ are calculated using Eqs. (41b) and (2).

5.4. Osterberg-Cell tests by Nam et al.⁴⁹

Nam et al.⁴⁹ conducted Osterberg-Cell tests on three shafts to investigate the effect of drilling tools on the ultimate side shear resistance of rock socketed shafts. The engineering properties of the rock

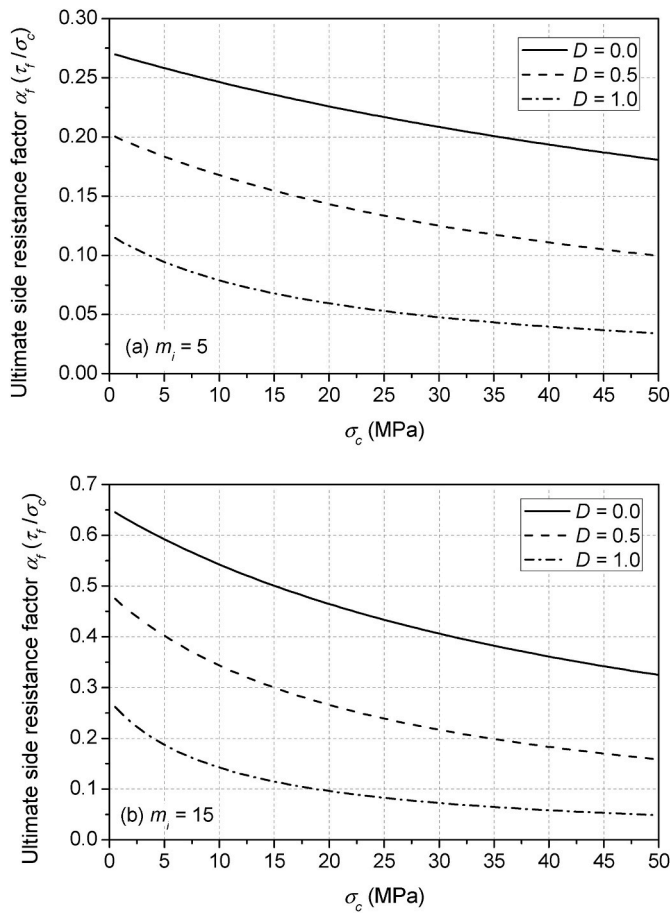


Fig. 10. Ultimate side shear resistance factor α_f versus σ_c : (a) $m_i = 5$; and (b) $m_i = 15$.

masses, σ_c , E_{rm} and RQD, are shown in Table 2. The m_i is again determined following Hoek and Brown⁴⁴ based on the rock type. Using the RQD values of 95, 82 and 97 for shafts #22, #23 and #24 and assuming an average J_{cond99} of 20, the GSI values are calculated using Eq. (42). The dimensions of each shaft are also shown in Table 2. A borehole roughness profile device was used to measure the roughness of each borehole during drilling, and the measured λ and h values of each shaft are summarized in Table 2. The equivalent asperity angle δ of each shaft is determined using Eq. (2).

5.5. Prediction results

With the engineering properties of rock masses, shaft dimensions, and sidewall roughness parameters, the τ_{sf} was determined for each shaft using the proposed method, the empirical equation of Xu et al.,¹⁵ and the predicted values are shown in Table 2. Since the construction-induced disturbance would affect the rock mass and thus the ultimate side shear resistance, it is necessary to take the disturbance factor into account. As no specific information regarding the degree of disturbance is available for the 24 cases, three different D values, 0.0 (no disturbance), 0.5 (medium disturbance), and 1.0 (severe disturbance), are considered in the analyses. Fig. 8 plots the predicted τ_{sf} from the proposed solution using different D values, the predictions by the empirical equation of Xu et al.¹⁵ as given in Table 1 and the measured τ_{sf} for the 24 rock socketed shafts. Among those shafts, shafts #1 to #5 are model test shafts while shafts #6 to #24 are field test shafts. For the five model tests, since they were installed in the mortar before curing, the construction disturbance should be negligible. Therefore, all the measured τ_{sf} values of the model shafts are close to the upper bound line

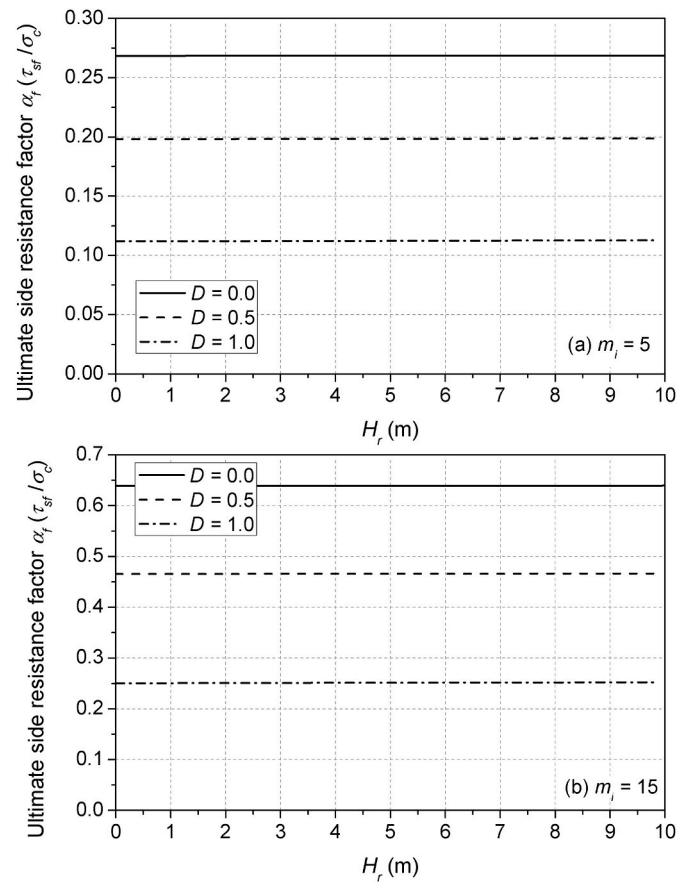


Fig. 11. Ultimate side shear resistance factor α_f versus H_r : (a) $m_i = 5$; and (b) $m_i = 15$.

of the proposed solution at $D = 0.0$. In contrast, the measured τ_{sf} values of almost all the field test shafts (except for shafts #10 and #23) are close to the average line of the proposed solution at $D = 0.5$. This means that the construction of rock socketed shafts in the field always causes some disturbance to the rock mass. Overall, it can be clearly seen that the predicted τ_{sf} values using the proposed method are in good agreement with the measured τ_{sf} values for both the model and field test shafts. Also, the disturbance factor D has a great effect on the predicted τ_{sf} values and should be properly determined by considering the disturbance of the rock mass from the construction process. Due to the ignorance of the sidewall roughness and rock types, the empirical equation of Xu et al.¹⁵ generally underestimates the τ_{sf} of shafts.

6. Parametric analyses

In this section, parametric analyses were conducted with the proposed method to investigate the effects of rock mass properties (GSI and σ_c), shaft dimensions (H_r and B), and roughness parameters (λ , δ , and RF) on the ultimate side shear resistance of rock socketed shafts. To better understand the behavior of rock socketed shafts related to rock mass type and construction method, two m_i values (5 for weak rock and 15 for strong rock) and three different D values (0.0, 0.5 and 1.0 for different degrees of construction disturbance) are considered. The basic parameters for the rock masses, shaft dimensions and sidewall roughness are listed in Table 5, and the parametric analyses are performed by changing one of those parameters at a time. To be simple, the predicted τ_{sf} is normalized by σ_c as the ultimate side shear resistance factor α_f in the analyses.

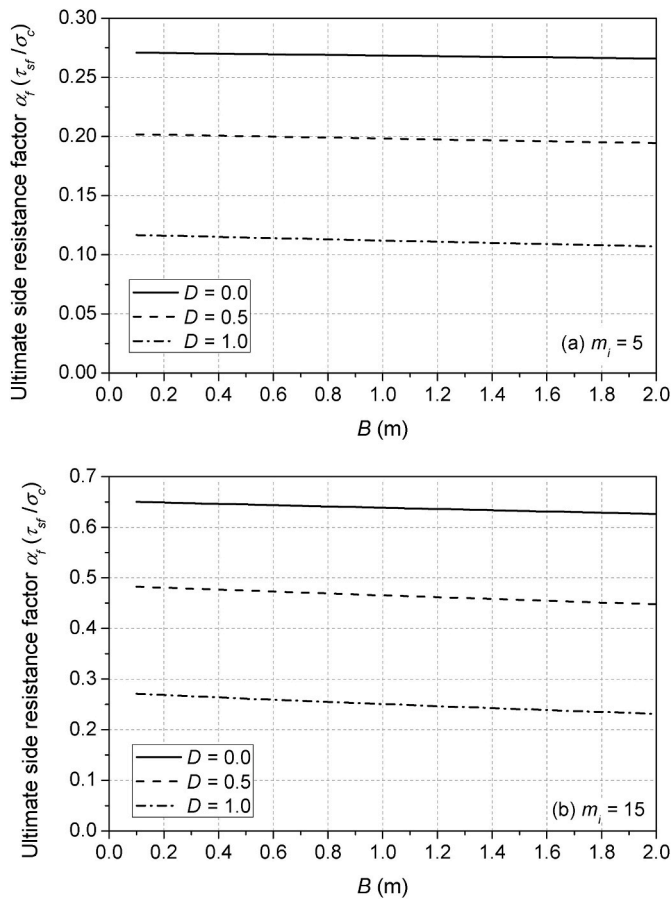


Fig. 12. Ultimate side shear resistance factor α_f versus B : (a) $m_i = 5$; and (b) $m_i = 15$

6.1. Effect of rock mass properties (GSI and σ_c)

Fig. 9 shows the α_f versus GSI for shafts socketed in rock masses with $m_i = 5$ and 15, respectively. Higher α_f is obtained for rock masses with higher m_i at the same GSI and D values. For different m_i and D values, the α_f significantly increases with GSI. At both $m_i = 5$ and 15, D barely affects the α_f when GSI is close to 100 or 0 because the effect of D on the rock mass parameters is negligible at GSI = 100 and 0 as indicated by Eqs. (4f)-(4h). For $20 < \text{GSI} < 95$, the α_f drastically decreases when D increases from 0 to 1. Therefore, for general rock mass conditions with $20 < \text{GSI} < 95$, it is important to reduce construction disturbance to the surrounding rock mass so as to ensure the high side shear resistance of rock socketed shafts.

The effect of unconfined compressive strength σ_c on the α_f is shown in Fig. 10. As expected, the α_f at $m_i = 15$ is much higher than that at $m_i = 5$ when the D and σ_c values are the same. At both m_i values and the different D values, the α_f decreases with higher σ_c and the effect of D is significant at all σ_c values.

6.2. Effect of shaft dimensions (H_r and B)

Fig. 11 shows the α_f versus H_r with $m_i = 5$ and 15 and different D values. For all the cases, the α_f barely changes with H_r . This is because the initial stress, which is mainly controlled by H_r , is much smaller compared to the failure stress. Again, the α_f at $m_i = 5$ is smaller than that at $m_i = 15$ at the same H_r and D values, and the disturbance factor D has a great effect on the α_f at all H_r and m_i values.

Fig. 12 plots the α_f versus the shaft diameter B with different m_i and D values. Since B cannot be zero, the curves in Fig. 12 all start at $B = 0.1$

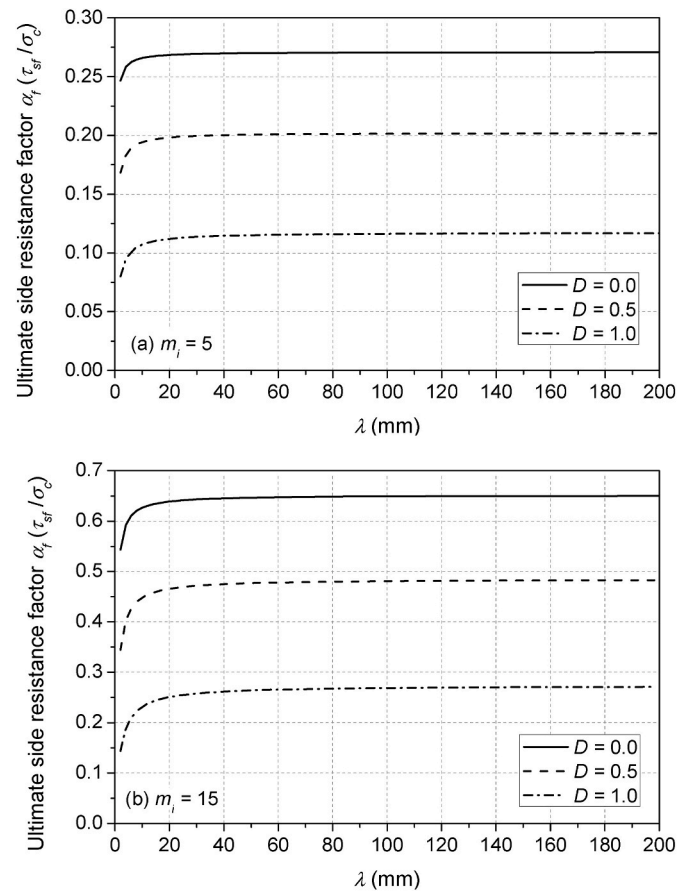


Fig. 13. Effect of λ on ultimate side shear resistance factor α_f : (a) $m_i = 5$; and (b) $m_i = 15$

m. At the same D and B values, the α_f at $m_i = 15$ is much larger than that at $m_i = 5$. The α_f slightly decreases with B for all m_i and D values. This is because the increase of B leads to the decrease of RF as indicated by Eq. (40). Again, the effect of D on the α_f is significant under all m_i and B values.

6.3. Effect of roughness parameters (λ , δ and RF)

Fig. 13 shows the effect of the asperity half chord length λ on the α_f with $m_i = 5$ and 15 and different D values. As expected, the α_f with $m_i = 15$ is much larger than that with $m_i = 5$ at the same λ and D values, and the disturbance factor D has a significant effect on the α_f at all λ and m_i values. It can also be seen that the α_f increases significantly with λ when λ is small, with the increasing rate declining with λ . When $\lambda > 100$ mm (1/10 of the diameter), the α_f barely changes with λ . This is because the shaft sidewall can be taken as smooth if λ (the size of asperities) is too small, and the ratio of $\frac{S_{sf}}{\lambda}$ tends to be constant when λ is large enough as indicated by Eq. (37a), which results in a constant τ_{sf} as shown in Eq. (38b).

Fig. 14 shows the effect of δ on α_f with $m_i = 5$ and 15 and different D values. At the same D and δ values, the α_f at $m_i = 15$ is much larger than that at $m_i = 5$. At all m_i and D values, the α_f first increases and then decreases with δ and stops at a critical asperity angle δ_{cr} , which is in agreement with the model test results by Dai et al.³¹ A possible explanation for the decrease of the α_f with δ is that the failure surface is shortened with the increase of δ and thus the rock asperity is more susceptible to shear failure after the δ exceeds a limiting value. Since this limiting δ corresponds to the maximum α_f , it is defined as the optimum asperity angle δ_{op} as shown in Fig. 14. It is noted that the δ_{op} increases

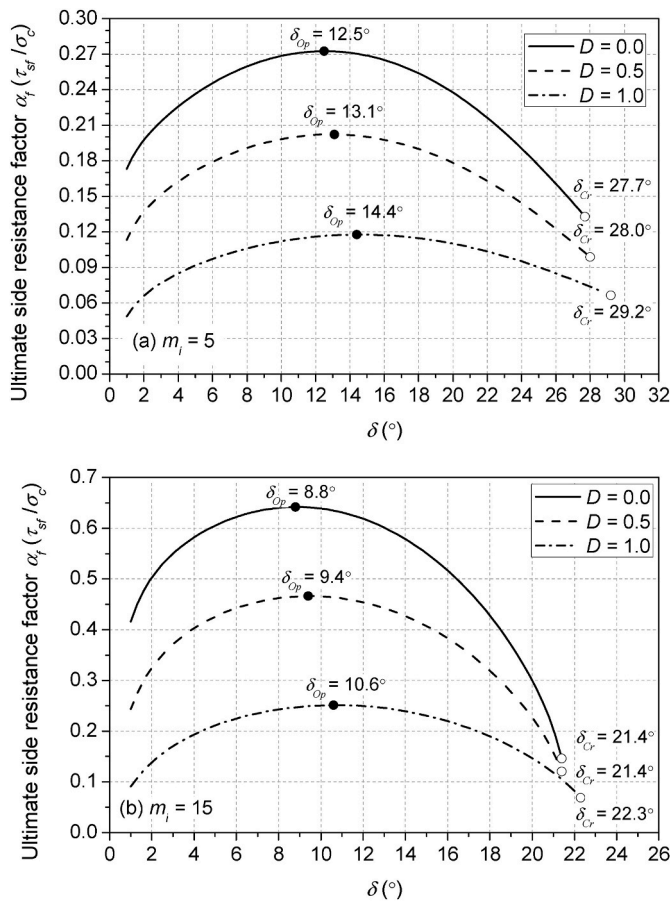


Fig. 14. Effect of δ on ultimate side shear resistance factor α_f : (a) $m_i = 5$; and (b) $m_i = 15$

with D but decreases with m_i . Therefore, it is recommended to use a smaller δ in rock masses with higher m_i and lower D . To get the maximum ultimate side shear resistance without using too large a δ (too high an RF value), a δ equal to the δ_{opt} (which is between 14.4° and 8.8° for m_i between 5 and 15) should be used. The critical angle δ_{cr} corresponds to the state when the log-spiral zone in Fig. 4 vanishes and the two constant state zones merge into one constant state zone. This means that further increase of δ beyond δ_{cr} would not increase the ultimate side shear resistance. It is interesting to note that the δ_{cr} increases with D but decreases with m_i .

As shown in Eq. (40), RF is related to both λ and δ . In this regard, Fig. 15 shows the effect of RF on the α_f with $m_i = 5$ and different D values by individually changing λ and δ , respectively. Similar trends can also be observed when $m_i = 15$. As expected, with a constant δ and different λ values, the effect of RF on α_f is similar to that of λ on α_f . If λ is kept as a constant and only δ changes, the effect of RF on α_f is similar to that of δ on RF. This means that the RF is not an independent factor and should be used with caution in analyzing the ultimate side shear resistance of rock socketed shafts.

7. Conclusion

This study proposes a new analytical solution for determining the ultimate side shear resistance of rock socketed shafts by considering 3D strength and sidewall roughness. The following conclusions can be made based on the present study:

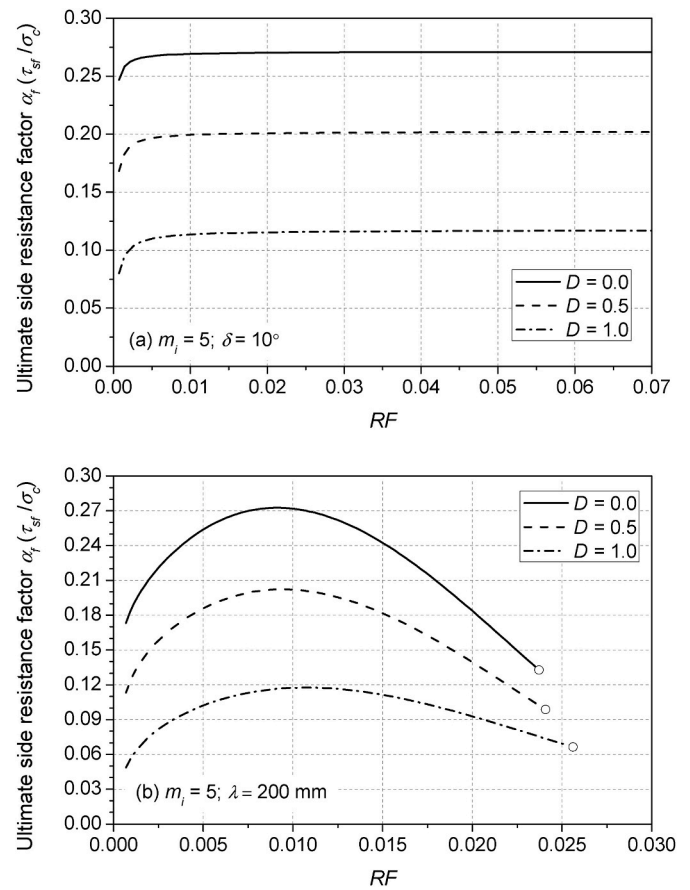


Fig. 15. Effect of RF on ultimate side shear resistance factor α_f with $m_i = 5$: (a) $\delta = 10^\circ$ and at different λ values; and (b) $\lambda = 200$ mm and at different δ values.

1. The proposed solution considering 3D strength and sidewall roughness makes reasonable predictions on the ultimate side shear resistance τ_{sf} of rock socketed shafts.
2. The disturbance factor D has a significant effect on the ultimate side shear resistance τ_{sf} of rock socketed shafts. At GSI = 75, when D is increased from 0 to 1, the ultimate side shear resistance τ_{sf} could drop by 50%. And the effect of D is more significant for rock mass with a smaller GSI.
3. The ultimate side shear resistance factor α_f of rock socketed shafts increases with m_i , GSI and asperity half chord length λ , but decreases with shaft diameter B and σ_c . Besides, the ultimate side shear resistance factor α_f first increases and then decreases with the asperity inclination angle δ . So, it is important to consider the various factors when designing a rock socketed shaft.
4. The RF is related to both λ and δ and thus not an independent factor when analyzing the ultimate side shear resistance of rock socketed shafts. To effectively improve the ultimate side shear resistance, an asperity half chord length λ of 1/10 of the diameter and an asperity inclination angle δ close to its optimum value δ_{opt} should be used.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Zhang L. *Drilled Shafts in Rock: Analysis and Design*. first ed. London, UK: A A Balkema; 2004.
- NCHRP Synthesis 360. *Rock - Socketed Shafts for Highway Structures Foundations*. Washington, D.C.: Transportation Research Board; 2006.
- Williams A, Donald I, Chiu H. Stress distributions in rock socketed piles. In: *Proceedings of the International Conference on Structural Foundations on Rock, Sydney, Australia*. 1980:317–326.
- Kaderabek TJ, Reynolds RT. Miami limestone foundation design and construction. *J. Geotech. Eng.* 1981;107(7):859–872.
- Reese LC, O'Neill MW. *Drilled Shafts: Construction Procedures and Design Methods. Design Manual*. Mclean, VA: U.S. Department of Transportation, Federal Highway Administration; 1987.
- Rosenberg P, Journeaux NL. Friction and end bearing tests on bedrock for high capacity socket design. *Can Geotech J.* 1976;13(3):324–333. <https://doi.org/10.1139/t76-033>.
- Horvath RG, Kenney TC. *Shaft Resistance of Rock Socketed Drilled Piers*. USA: *Proc Am Soc Civil Eng Annu Conv*. Atlanta; 1979:182–214.
- Meigh AC, Wolski W. Design parameters for weak rocks. In: *Design Parameters in Geotechnical Engineering Proc 7th European Conference on Soil Mechanics and Foundation Engineering Brighton, 1979*. vol. 5. London: British Geotechnical Society; distributed by Telford Ltd; 1981:59–79.
- Williams AF. *The Design and Performance of Piles Socketed into Weak Rock*. PhD Dissertation. Melbourne, Australia: Monash University; 1980.
- Horvath RG. *Drilled Piers Socketed into Weak Shale-Methods of Improving Performance*. PhD Dissertation. Toronto, Ontario, Canada: University of Toronto; 1982.
- Rowe R, Armitage H. *The Design of Piles Socketed into Weak Rock*. London, ON, Canada: Faculty of Engineering Science, University of Western Ontario; 1984.
- Kulhawy FH, Phoon KK. Drilled shaft side resistance in clay soil to rock. In: *Design and Performance of Deep Foundations. Piles and Piers in Soil and Soft Rock*. Geotechnical Special Publication No. 38, ASCE; 1993:172–183.
- Carrubba P. Skin friction on large-diameter piles socketed into rock. *Can Geotech J.* 1997;34(2):230–240. <https://doi.org/10.1139/t96-104>.
- Zhang LY, Einstein HH. End bearing capacity of drilled shafts in rock. *J Geotech Geoenviron Eng.* 1998;124(7):574–584. [https://doi.org/10.1061/\(asce\)1090-0241\(1998\)124:7\(574\)](https://doi.org/10.1061/(asce)1090-0241(1998)124:7(574)).
- Xu J, Gong W, Gamage RP, Zhang Q, Dai G. A new method for predicting the ultimate shaft resistance of rock-socketed drilled shafts. *P I Civil Eng Geotech.* 2020; 173(2):169–186. <https://doi.org/10.1680/jgeen.18.00221>.
- Williams AF, Pells PJN. Side resistance rock sockets in sandstone, mudstone and shale. *Can Geotech J.* 1981;18(4):505–513. <https://doi.org/10.1139/t81-061>.
- Johnston IW, Lam TSK, Williams AF. Constant normal stiffness direct shear testing for socketed pile design in weak rock. *Geotechnique*. 1987;37(1):83–89. <https://doi.org/10.1680/geot.1987.37.1.83>.
- Jiang Y, Xiao J, Tanabashi Y, Mizokami T. Development of an automated servo-controlled direct shear apparatus applying a constant normal stiffness condition. *Int J Rock Mech Min Sci.* 2004;41(2):275–286. <https://doi.org/10.1016/j.ijrmms.2003.08.004>.
- Indraratna B, Thirukumaran S, Brown ET, Zhu SP. Modelling the shear behaviour of rock joints with asperity damage under constant normal stiffness. *Rock Mech Rock Eng.* 2015;48:179–195. <https://doi.org/10.1007/s00603-014-0556-2>.
- Leong EC, Randolph MF. Finite element modeling of rock-socketed piles. *Int J Numer Anal Methods Geomech.* 1994;18(1):25–47.
- Chung JH, Ko J, Klammler H, McVay MC, Lai P. A numerical and experimental study of bearing stiffness of drilled shafts socketed in heterogeneous rock. *Comput Struct.* 2012;90–91:145–152. <https://doi.org/10.1016/j.compstruc.2011.09.004>.
- Gutiérrez-Ch JG, Senent S, Melentijevic S, Jimenez R. A DEM-based factor to design rock-socketed piles considering socket roughness. *Rock Mech Rock Eng.* 2021;54: 3409–3421. <https://doi.org/10.1007/s00603-020-02347-1>.
- Johnston IW, Lam TS. Shear behavior of regular triangular concrete/rock joints-analysis. *J Geotech Eng.* 1989;115(5):711–727. [https://doi.org/10.1061/\(ASCE\)0733-9410\(1989\)115:5\(711\)](https://doi.org/10.1061/(ASCE)0733-9410(1989)115:5(711)).
- Seidel JP, Haberfield CM. A theoretical model for rock joints subjected to constant normal stiffness direct shear. *Int J Rock Mech Min Sci.* 2002;39(5):539–553. [https://doi.org/10.1016/S1365-1609\(02\)00056-4](https://doi.org/10.1016/S1365-1609(02)00056-4).
- Sokolovskii VV. *Statics of Soil Media*. London, UK: Butterworth; 1960.
- Sokolovskii VV. *Statics of Granular Media*. London, UK: Pergamon Press; 1965.
- Zhao H, Xiao Y, Zhao M, Yin P. On behavior of load transfer for drilled shafts embedded in weak rocks. *Comput Geotech.* 2017;85:177–185. <https://doi.org/10.1016/j.compgeo.2016.12.023>.
- Zhang QQ, Ma B, Liu SW, Feng RF. Behaviour analysis on the vertically loaded bored pile socketed into weak rocks using slip-line theory arc failure surface. *Comput Geotech.* 2020, 103852. <https://doi.org/10.1016/j.compgeo.2020.103852>.
- Hou JC, Zhang H, Peng WZ, Zhao MH. A limit solution for predicting side resistance on rock-socketed piles. *J Eng Mech.* 2022;148(1), 04021131. [https://doi.org/10.1061/\(ASCE\)JEM.1943-7889.0002046](https://doi.org/10.1061/(ASCE)JEM.1943-7889.0002046).
- Serrano A, Olalla C. Shaft resistance of a pile embedded in rock. *Int J Rock Mech Min Sci.* 2004;41(1):21–35. [https://doi.org/10.1016/S1365-1609\(03\)00070-4](https://doi.org/10.1016/S1365-1609(03)00070-4).
- Dai GL, Gong WM, Chen L. Modified method for shaft resistance of rock-socketed piles based on Hoek-Brown criterion. *Chin J Geotech Eng.* 2012;34(9):1746–1752.
- Chen HH, Zhu HH, Zhang LY. A unified constitutive model for rock based on newly modified GZZ criterion. *Rock Mech Rock Eng.* 2021;54:921–935. <https://doi.org/10.1007/s00603-020-02293-y>.
- Chen HH, Zhu HH, Zhang LY. Analytical solution for deep circular tunnels in rock with consideration of disturbed zone, 3D strength and large strain. *Rock Mech Rock Eng.* 2021;54:1391–1410. <https://doi.org/10.1007/s00603-020-02339-1>.
- Potyondy JG. Skin friction between various soils and construction materials. *Geotechnique*. 1961;11:339–353. <https://doi.org/10.1680/geot.1961.11.4.339>.
- Hoek E, Diederichs MS. Empirical estimation of rock mass modulus. *Int J Rock Mech Min Sci.* 2006;43:203–215. <https://doi.org/10.1016/j.ijrmms.2005.06.005>.
- Chen HH, Zhu HH, Zhang LY. A three-dimensional (3D) analytical solution for the ultimate end-bearing capacity of rock-socketed shafts. *Rock Mech Rock Eng.* 2022;55: 611–627. <https://doi.org/10.1007/s00603-021-02710-w>.
- Myers NO. Characterization of surface roughness. *Wear.* 1962;5(3):182–189.
- Tse R, Cruden DM. Estimating joint roughness coefficients. *Int J Rock Mech Min Sci Geomech Abstr.* 1979;16(5):303–307. [https://doi.org/10.1016/0148-9062\(79\)90241-9](https://doi.org/10.1016/0148-9062(79)90241-9).
- Horvath RG, Kenney TC, Trow WA. Results of tests to determine shaft resistance of rock socketed drilled piers. In: *Proceedings of International Conference on Structural Foundations on Rock, Sydney*. 1980:349–361.
- Pells PJN, Rowe RK, Turner RM. An experimental investigation into side shear for socketed piles in sandstone. In: *Proceedings of the International Conference on Structural Foundations on Rock, Sydney*. 1980:291–302.
- Park BG, Nam MS, Choi Y. Development and field verification of borehole roughness profiler for drilled shafts. In: Hamza M, et al., eds. *Proceedings of the 17th International Conference on Soil Mechanics and Geotechnical Engineering*. 2009:1139–1142. <https://doi.org/10.3233/978-1-60750-031-5-1139>.
- Ashtoto. *LRFD Bridge Design Specifications*. seventh ed. Washington, D.C.: American Association of State Highway and Transportation Officials; 2014.
- Dai GL, Salgado R, Gong WM, Zhu M. The effect of sidewall roughness on the shaft resistance of rock-socketed piles. *Acta Geotech.* 2017;12(2):429–440. <https://doi.org/10.1007/s11440-016-0470-8>.
- Hoek E, Brown ET. Practical estimates of rock mass strength. *Int J Rock Mech Min Sci Geomech Abstr.* 1997;34:1165–1186. [https://doi.org/10.1016/S1365-1609\(97\)80069-X](https://doi.org/10.1016/S1365-1609(97)80069-X).
- Williams AF, Pells PJN. Side resistance rock sockets in sandstone, mudstone, and shale. *Can Geotech J.* 1981;18(4):502–513. <https://doi.org/10.1139/t81-061>.
- Horvath RG, Kenney TC, Kozicki P. Methods for improving the performance of drilled piers in weak rock. *Can Geotech J.* 1983;20(4):758–772. <https://doi.org/10.1139/t83-081>.
- Hoek E, Carter TG, Diederichs MS. Quantification of the geological strength index chart. In: *47th US Rock Mech./Geomech. Symp., San Francisco, CA, USA*. 2013:13–672.
- Bieniawski. *Engineering Rock Mass Classifications*. Rotterdam: John Wiley; 1989.
- Nam MS, Vipulanandan C. Roughness and unit side resistance of drilled shafts socketed in clay shale and limestone. *J Geotech Geoenviron Eng.* 2008;134(9): 1272–1279. [https://doi.org/10.1061/\(ASCE\)1090-0241\(2008\)134:9\(1272\)](https://doi.org/10.1061/(ASCE)1090-0241(2008)134:9(1272)).